

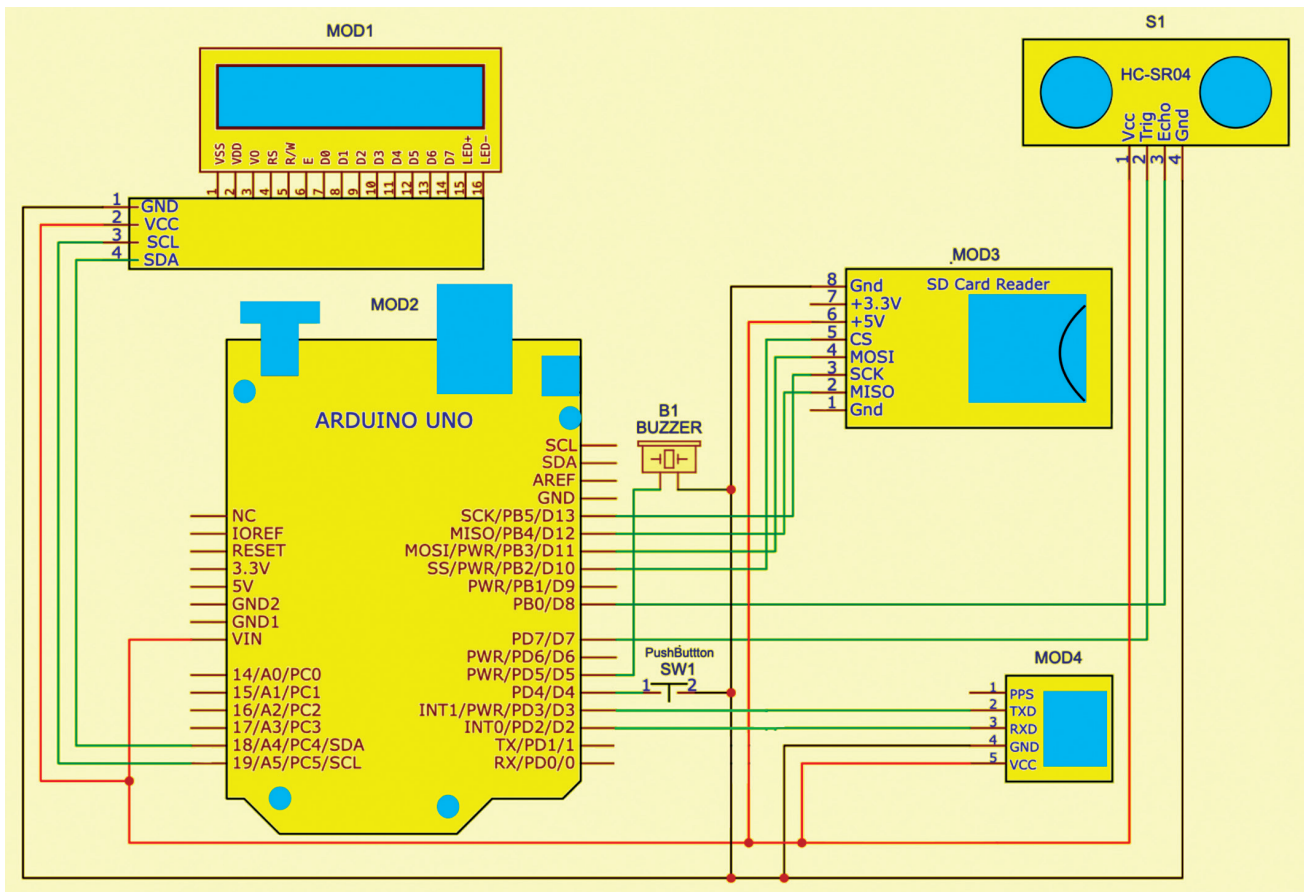


Fig. 2: Circuit diagram

distance measurement.

The GPS module is interfaced via the SoftwareSerial pins 2 (RX) and 3 (TX), providing real-time latitude and longitude data to geotag each measurement.

The SD card module is connected through the SPI interface. The Arduino Uno pins D10, D11, D12, and D13 are connected to the SD card reader's CS, MOSI, MISO, and SCK pins, allowing the storage of depth and positional data in CSV format for post-processing, mapping, and documentation.

A 16×2 LCD is connected via the I²C interface (SDA and SCL) to display real-time system parameters, including measured depth, GPS status, and logging confirmation. A push-button switch (SW1), connected to pin 4, is used to manually trigger data recording, while a buzzer (B1) on pin 5 provides audible feedback to confirm

successful data capture.

The entire system is powered by a regulated 5V supply, typically from a USB power bank connected to the Arduino, which in turn distributes power to all modules.

Overall, the integrated system operates as a synchronised and automated data logger, combining depth measurement with location tagging. This significantly improves efficiency, measurement accuracy, and documentation in OFC trench deployment operations.

Software

The software for the portable ultrasonic trench depth and location logger system is developed using the Arduino IDE and integrates multiple modules, including the ultrasonic sensor, GPS receiver, SD card, LCD, buzzer, and push button. It enables automated depth measurement and geolocation logging

by synchronising sensor data with GPS coordinates and data storage, ensuring reliable and accurate results.

The program is designed to be user-friendly, with well-structured code and clear comments. To simplify development and handle complex tasks efficiently, several libraries are used. The Wire.h and LiquidCrystal_I2C.h libraries support I²C communication and LCD display control. The TinyGPSPlus.h library decodes GPS data into readable coordinates, while SoftwareSerial.h enables serial communication with the GPS module.

For data storage, SPI.h and SD.h libraries are used to interface with the SD card and perform file operations. Additionally, TimeLib.h can be included for time-related functions, if required. Fig. 3 shows the code snippet with the included libraries.